THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** three lakh forty-five thousand six hundred seventy-eight	Read period by period: crore, lakh, thousand.
2	**(a)** 26	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
3	**(a)** 0.06	Find the place of the digit: tenths, hundredths or thousandths.
4	**(c)** 11	Replace x by its value and calculate.
5	**(b)** 60°	Complementary angles add up to 90°.
6	**(d)** even	Use the even/odd rules.
7	**(b)** Yes	Two sides together must be longer than the third.
8	**(c)** $\frac{2}{5}$	Multiply numerators and multiply denominators, then simplify.
9	**(b)** 2,53,400	Place each period.
10	**(c)** 42	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
11	**(c)** 6.1	Line up the decimal points and add.
12	**(c)** 11	Replace y by its value and subtract.
13	**(a)** 130°	Supplementary angles add up to 180°.
14	**(b)** 11	Add all the digits.
15	**(d)** 70°	The three angles add up to 180°.
16	**(d)** 8	'Of' means multiply.
17	**(a)** 4,73,000	Add and re-insert commas.
18	**(d)** 6	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
19	**(a)** 5.3	Line up the decimal points and subtract.
20	**(a)** $n + 5$	Translate the words into a letter-number expression.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Read period by period (1); three lakh forty-five thousand six hundred (1).	1+1
22	Multiply/divide first; Then add/subtract (1); Value $= 33$ (1).	1+1
23	Write the numbers with the decimal points one below the other (1); Sum $= 17.15$ (1).	1+1
24	Substitute: $2(3) + 5$ (1); $= 11$ (1).	1+1
25	Complement $= 90^\circ - 35^\circ = 55^\circ$ (1); Supplement $= 180^\circ - 110^\circ = 70^\circ$ (1).	1+1
26	Apply the rule to each number in turn; Sequence: $6, 3, 10, 5, 16, 8, 4, 2, 1$ (1); It reaches 1 after 8 steps (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	International: 345,678 (1); Expanded: 300000 + 40000 + 5000 + 600 + 70 + 8 (2).	1+2
28	Brackets first; Then multiplication/division (1); Then addition/subtraction (1); Value $= 50$ (1).	1+1+1
29	$0.6 = \frac{6}{10}$ (1); Divide numerator and denominator by their common factor (1); Simplest form: $\frac{3}{5}$ (1).	1+1+1
30	Substitute: $2(4) + 3(5)$ (1); $= 8 + 15$ (1); $= 23$ (1).	1+1+1
31	Sum $= 180^\circ$ and the number of parts $= 3$ (1); One part $= 60^\circ$ (1); The angles are 60° and 120° (1).	1+1+1
	OR: Largest $= 5432$ (1); Smallest $= 2345$ (digits arranged in increasing order) (1); Difference $= 5432 - 2345 = 3087$ (1).	1+1+1
32	Total parts $= 6$; one part $= 180^\circ \div 6 = 30^\circ$ (1); Angles 30°, 60°, 90° (1); Check: 180° (1).	1+1+1
	OR: Improper form: $2\frac{1}{3} = \frac{7}{3}$ (1); $\frac{7}{3} \times \frac{3}{5} = \frac{21}{15}$ (1); Answer $1\frac{2}{5}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Total $= 12,50,000 + 9,80,000 = 22,30,000$ (1); Difference $= 2,70,000$ (1); Total in words: twenty-two lakh thirty thousand (1); State units (rupees) (2).	1+1+1+2
34	Total $= 6 \times 5 + 4$ (1); $= 34$ (1); Present $= 34 - 3$ (1); $= 31$ (1); Check the order of operations (1).	1+1+1+1+1
35	Left $= 5.75 - 2.4 = 3.35$ m (2); In centimetres: 335 cm (3).	2+3
36	Sister's age $= x - 3$ (1); Sum $= x + (x - 3) = 2x - 3$ (1); At $x = 15$: sister $= 12$ years (1); Sum $= 27$ years (2).	1+1+1+2
	OR: Vertically opposite angle $= 35^\circ$ (1); The other two angles are 145° each (linear pair) (1); Sum $= 35 + 35 + 145 + 145 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
37	Reverse $= 62$ (1); Sum $= 26 + 62 = 88$ (1); $88 = 11 \times 8$ (1); Divisible by 11; quotient $= 8$ (2).	1+1+1+2
	OR: $5 + 5 = 10 > 8$, so the triangle exists (1); It is isosceles (1); Perimeter $= 5 + 5 + 8$ (1); $= 18$ cm (1); Final statement (1).	1+1+1+1+1
38	At first $\frac{3}{4} \times 120 = 90$ L (1); Used $\frac{2}{3} \times 90 = 60$ L (1); Left $= 90 - 60$ (1); $= 30$ L (1); State all three answers (1).	1+1+1+1+1
	OR: $25,000 \times 12 = 3,00,000$ (2); In words: three lakh (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 40°	Vertically opposite angles are equal.
2	**(d)** 5743	Write the digits in the opposite order.
3	**(d)** equilateral	Recall the definition.
4	**(c)** $\frac{7}{3}$	Turn the fraction upside down.
5	**(b)** 3,76,544	Subtract place by place.
6	**(d)** $(7+5)\times 3$	Brackets show the sum is done first.
7	**(c)** 45.2	Multiplying by 10 moves the decimal point 1 place(s) to the right.
8	**(a)** 3	Terms are separated by + or - signs.
9	**(c)** 120°	A linear pair adds up to 180°.
10	**(d)** 1331	A palindrome reads the same from both ends.
11	**(b)** 95°	Exterior angle = sum of the two interior opposite angles.
12	**(c)** $\frac{1}{4}$	Dividing by n means multiplying by $\frac{1}{n}$.
13	**(d)** 1,00,000	Indian place value system.
14	**(c)** $(20-8)\div 4$	Difference first, then divide.
15	**(c)** 4.56	Dividing by 10 moves the decimal point 1 place(s) to the left.
16	**(d)** 7	The coefficient is the number multiplying the letter.
17	**(c)** parallel lines	Recall the definition.
18	**(a)** 1827	A number is divisible by 9 if its digit sum is.
19	**(a)** 70°	Base angles are equal and the three angles add up to 180°.
20	**(a)** smaller than each of the fractions	Recall the idea from the chapter.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Two shorter sides: $6+8=14$; Compare with the longest side 10 (1); Yes, 14 > 10 (1).	1+1
22	$\frac{3}{5}\times\frac{10}{21}=\frac{3\times10}{5\times21};=\frac{30}{105}$ (1); Lowest terms: $\frac{2}{7}$ (1).	1+1
23	Place the periods (1); 6,30,500 (1).	1+1
24	Do the bracket first; Then the remaining operation (1); Value $=78$ (1).	1+1
25	Write the numbers with the decimal points one below the other (1); Difference $=11.75$ (1).	1+1
26	Identify the operations (1); Expression: $4n+7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Difference $=30,55,000$ (1); Nearest lakh: $31,00,000$ (2).	1+2
28	$3\times8+5$ (1); Value $=29$ (1); Check the order of operations (1).	1+1+1
29	Compare whole parts, then tenths, then hundredths (1); Ascending order: $0.055, 0.45, 0.5, 0.505$ (2).	1+2
30	All three terms are like terms (1); Coefficients: $7+4-3=8$ (1); Answer: $8x$ (1).	1+1+1
31	$x=180-110=70$ (1); The angle 70° is acute (2).	1+2
	OR: Middle number $=48\div3=16$ (1); The numbers are 15, 16 and 17 (1); Check: $15+16+17=48$ (1).	1+1+1
32	Base angles are equal; their sum $=180^\circ-30^\circ=150^\circ$ (1); Each base angle $=75^\circ$ (1); Check: the three angles add up to 180° (1).	1+1+1
	OR: Pieces $=n\div\frac{3}{4}$ (1); $6\times\frac{4}{3}=\frac{24}{3}$ (1); Pieces $=8$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	First $= 24$ (1); Second $= 36$ (1); Not equal; the second is greater (1); Brackets can change the value (2).	1+1+1+2
34	$1.25 \times 6 = 7.5$ kg (2); In grams: 7500 g (3).	2+3
35	Perimeter $= 2(l + 4)$ (1); Area $= 4l$ (1); At $l = 9$: perimeter $= 26$ cm (1); Area $= 36$ cm^2 (2).	1+1+1+2
36	Let the complement be y; the angle is $2y$ and $2y + y = 90$ (1); $y = 30$ (2); The angle is 60° and its complement is 30° (2).	1+2+2
	OR: Sequence: $9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Steps $= 19$ (1); Largest number $= 52$ (1); Even numbers: 13 (1); Ends in the loop 4, 2, 1 (1).	1+1+1+1+1
37	$\angle C = 180^\circ - 40^\circ - 65^\circ = 75^\circ$ (1); Classification: acute-angled (1); Exterior angle at C $= 180^\circ - 75^\circ$ (1); $= 105^\circ$ (1); It equals $\angle A + \angle B = 40^\circ + 65^\circ$ (1).	1+1+1+1+1
	OR: Cakes $= 5 \div \frac{3}{4} = 6\frac{2}{3}$ (1); Full cakes $= 6$ (1); Flour used $= 6 \times \frac{3}{4} = 4\frac{1}{2}$ kg (1); Left over $= \frac{1}{2}$ kg (1); Final statement (1).	1+1+1+1+1
38	Total $= 23,40,000 + 17,60,000 = 41,00,000$ (1); Difference $= 5,80,000$ (1); Total in words: forty-one lakh (1); State units (rupees) (2).	1+1+1+2
	OR: Total $= 8 \times 4 + 6$ (1); $= 38$ (1); Present $= 38 - 2$ (1); $= 36$ (1); Check the order of operations (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 48000	Look at the hundreds digit.
2	**(b)** 116	Multiply, then add.
3	**(c)** 0.7	Compare the whole parts, then tenths, then hundredths.
4	**(b)** $3x$ and $9x$	Like terms have the same letter part.
5	**(d)** 55°	Corresponding angles are equal when the lines are parallel.
6	**(a)** 22	Check whether the number is even or odd.
7	**(c)** 21	Add the three sides.
8	**(d)** $\frac{7}{3}$	Whole $\times$ denominator + numerator, over the denominator.
9	**(c)** 7	Count the digits.
10	**(d)** 50	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
11	**(c)** $\frac{1}{2}$	Write over 10, 100 or 1000 and reduce.
12	**(b)** 12	Add the two values.
13	**(b)** 48°	Alternate interior angles are equal when the lines are parallel.
14	**(b)** 36	The sum of the first n odd numbers is n^2.
15	**(c)** SSS	Compare the matching parts.
16	**(d)** $\frac{1}{3}$	Area = length $\times$ breadth.
17	**(d)** 1,00,000	Add 1.
18	**(a)** 17	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
19	**(d)** 0.7	The denominator tells the place: 10 tenths, 100 hundredths, 1000 thousandths.
20	**(b)** 20	Multiply the coefficient by the value of x.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Vertically opposite angle $= 65^\circ$ (1); The other two angles form a linear pair with it: $180^\circ - 65^\circ = 115^\circ$ each (1).	1+1
22	Add the number to its reverse; $29 + 92 = 121$ (1); Palindrome obtained: 121 (1).	1+1
23	Third $= 180^\circ - (48^\circ + 72^\circ)$; $= 60^\circ$ (1); The triangle is acute-angled (1).	1+1
24	Reciprocal of $\frac{9}{16}$ is $\frac{16}{9}$; $\frac{3}{4} \times \frac{16}{9} = \frac{48}{36}$ (1); Answer $1\frac{1}{3}$ (1).	1+1
25	Add column by column (1); $4, 22, 100$ (1).	1+1
26	$7 \times 8 + 9$ (1); $= 65$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Words: two lakh thirty-four thousand five hundred sixty-seven (1); Nearest ten thousand: $2,30,000$ (2).	1+2
28	$6 \times 5 + 4 \times 3$ (1); $30 + 12$ (1); $= 42$ (1).	1+1+1
29	Look at the digit just after the place being rounded (1); (i) 6.3 (1); (ii) 6 (1).	1+1+1
30	Cost $= 5p$ (1); At $p = 12$: 5×12 (1); Cost $=$ ₹60 (1).	1+1+1
31	(i) Alternate interior angles are equal: 52° (1); (ii) Interior angles on the same side add up to 180°: 128° (2).	1+2
	OR: 13×7 is odd (odd $\times$ odd) (1); 4 is even (1); The sum is odd (1).	1+1+1
32	Exterior $=$ sum of interior opposite angles (1); Other $= 120^\circ - 45^\circ = 75^\circ$ (1); Adjacent interior angle $= 180^\circ - 120^\circ = 60^\circ$ (1).	1+1+1
	OR: Area $= \frac{3}{4} \times \frac{2}{3}$; Area $= \frac{1}{2}$ m^2 (1); Perimeter $= 2(\frac{3}{4} + \frac{2}{3})$; $= 2 \times \frac{17}{12}$ (1); Perimeter $= 2\frac{5}{6}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Left $= 8.5 - 3.65 = 4.85$ m (2); In centimetres: 485 cm (3).	2+3
34	Sister's age $= x - 5$ (1); Sum $= x + (x - 5) = 2x - 5$ (1); At $x = 18$: sister $= 13$ years (1); Sum $= 31$ years (2).	1+1+1+2
35	Vertically opposite angle $= 50^\circ$ (1); The other two angles are 130° each (linear pair) (1); Sum $= 50 + 50 + 130 + 130 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
36	Reverse $= 74$ (1); Sum $= 47 + 74 = 121$ (1); $121 = 11 \times 11$ (1); Divisible by 11; quotient $= 11$ (2).	1+1+1+2
	OR: $6 + 8 = 14 > 10$, so the triangle exists (1); It is scalene (1); Perimeter $= 6 + 8 + 10$ (1); $= 24$ cm (1); Final statement (1).	1+1+1+1+1
37	At first $\frac{5}{8} \times 80 = 50$ L (1); Used $\frac{4}{5} \times 50 = 40$ L (1); Left $= 50 - 40$ (1); $= 10$ L (1); State all three answers (1).	1+1+1+1+1
	OR: $1,50,000 \times 8 = 12,00,000$ (2); In words: twelve lakh (3).	2+3
38	First $= 28$ (1); Second $= 4$ (1); Not equal; the first is greater (1); Brackets can change the value (2).	1+1+1+2
	OR: $0.75 \times 8 = 6$ kg (2); In grams: 6000 g (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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